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EE 472

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Homework #2

Implementation of a full adder using dataflow modeling.

The screenshot displays the Quartus Prime IDE interface. The Project Navigator on the left shows the hierarchy of the project, including the MAX 10: 10M08DAF484C8G device and the full_adder module. The Tasks window shows the compilation process, with the following tasks listed:

Task
Compile Design
Analysis & Synthesis
Fitter (Place & Route)
Assembler (Generate program)

The main window displays the Verilog code for the full_adder module. The code is as follows:

```
1 //A Full Adder is composed of two xor gates, two
2 module full_adder
3 (
4     output wire S, C_out,
5     input wire A, B, C_in
6 );
7 wire XOR1_out, AND1_out, AND2_out;
8 assign XOR1_out = A ^ B;
9 assign AND1_out = XOR1_out && C_in;
10 assign AND2_out = A && B;
11 assign S = XOR1_out ^ C_in;
12 assign C_out = AND1_out || AND2_out;
13 endmodule
14
```

The bottom status bar shows the compilation results: 293000 Quartus Prime Full Compilation was successful. 0 errors, 13 warnings.